AQUA-SENSE: AI-POWERED IOT SYSTEM FOR PUBLIC WATER QUALITY MONITORING & PREDICTIVE ALERTS

FEASIBILITY STUDY

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LITERATURE SURVEY

The papers reviewed highlight major advancements in the integration of IoT, sensor technologies, machine learning, and cloud computing within the domain of water quality monitoring. These studies demonstrate how real-time data acquisition, wireless communication, and intelligent data analysis are transforming traditional water testing methods into intelligent, proactive, and accessible solutions. However, most of the works focus on specific technical aspects—such as a particular sensing modality, communication protocol, or data analysis technique—rather than offering a unified, scalable, and cost-effective system adaptable to diverse aquatic environments. The following section provides an overview of related research papers that guided the development of our proposed AQUA-SENSE solution.

1. The paper "IoT-Enabled Advanced Water Quality Monitoring System for Pond Management and Environmental Conservation" focuses on a mobile sensing platform for spatial data collection. The study demonstrates that an ESP32-based system deployed on an aquatic boat can provide real-time monitoring of pH, turbidity, and TDS across different points in a pond, improving data accuracy over single-point sampling.

2. The paper "A Low-Cost LoRa Optical Fluorometer Nephelometer for Wireless Monitoring of Water Quality Parameters in Real Time" explores the use of optical sensors and long-range, low-power communication. It introduces a portable, open-source device that measures biological indicators like chlorophyll-a, offering a cost-effective method for detecting algal blooms in field deployments.

3. In the paper "A Hybrid Machine Learning and Embedded IoT-based Water Quality Monitoring System," the authors present a comprehensive model that integrates multiple sensors with cloud-based Artificial Neural Networks (ANN) and Support Vector Machines (SVM). The system provides high-accuracy water quality predictions and enables automated corrective treatment, showcasing the potential of AI in proactive water management.

4. The study “Connected Sensors, Innovative Sensor Deployment, and Intelligent Data Analysis for Online Water Quality Monitoring” proposes a multi-sensor network deployed on buoys, robots, and drones. The system provides broad spatiotemporal coverage and uses AI for standardized analysis and forecasting, enabling the early detection of contaminants over wide areas.

5. The paper “Toward Design of IoT and Machine Learning-Enabled Frameworks for Analysis and Prediction of Water Quality” discusses the use of regression and classification models to compute Water Quality Index (WQI) and Class (WQC). It ensures continuous, real-time assessment, making advanced water quality evaluation more accessible compared to infrequent lab testing.

6. In “Smart system for water quality monitoring utilizing long-range IoT,” the authors propose a model that utilizes LoRa technology for communication. The approach achieves high accuracy in measuring pH and turbidity with minimal power consumption, demonstrating effective long-range monitoring for remote locations.

7. The paper “IoT Real-Time Potable Water Quality Monitoring and Prediction” introduces a system that combines multi-parameter sensing with NB-IoT connectivity and machine learning. It achieves high prediction accuracy and provides real-time alerts via web dashboards and mobile apps, enhancing the reliability of potable water monitoring.

8. In “AQUASENSE – Sensor-based Water Quality Monitoring Device,” the study highlights a low-cost, Arduino-based solution for basic water quality assessment. The system provides real-time results for TDS, pH, and temperature, offering a portable and efficient alternative to commercial monitors for initial water screening.

9. The paper “IoT-Based Water Quality Monitoring in Philippine Off-Grid Communities” presents a solar-powered, community-centric system designed for

rural areas. It integrates GSM for SMS alerts, ensuring that critical water quality information reaches authorities even in regions with limited internet infrastructure.

10. The paper “Wireless Sensor Network IoT Platform for Water Consumption and Quality” proposes a multi-user platform that monitors both water quality and consumption. The system uses an ESP32-based sensor network and custom applications for different stakeholders, providing a cost-effective and holistic view of water usage and safety.

Overall, the literature reviewed demonstrates the wide range of technologies applied in the development of modern water quality monitoring systems. While existing studies have achieved notable success in enhancing sensing accuracy, data transmission range, and predictive capabilities, there remains scope for improvement in integrating these approaches into a single, efficient, and scalable solution.

NEEDS ASSESSMENT

Target Audience

The Aqua-sense Water Quality Monitoring System is designed to address the critical need for safe and transparent water management for a wide range of stakeholders. The system primarily targets:

- Public Water Supply Authorities and Municipal Corporations: Entities responsible for maintaining the quality and safety of public water distribution networks. They require timely, accurate data to identify contamination events, manage resources, and uphold public health standards.
- Administrative and Regulatory Bodies: Organizations that need access to historical and real-time water quality data for compliance monitoring, auditing, and informed policy- making.
- Consumers and the General Public: End-users of the water supply who seek assurance about the safety and quality of the water they consume daily. A transparent system fosters public trust and allows for direct feedback.
- Private Tank and Society Management: Residential complexes and industries that rely on private water storage tanks can use the system to monitor their local water quality, ensuring the integrity of their internal water systems.

Identified Needs

Access to clean and safe drinking water is a fundamental necessity, yet traditional water quality monitoring methods are often slow, manual, and opaque. Current practices typically involve periodic physical sample collection from various points in the distribution network, followed by laboratory analysis. This process is time-consuming, expensive, and provides only a sporadic snapshot of water quality, leaving large gaps where contamination events can go undetected for extended periods.

This latency in detection can lead to public health crises, erosion of trust in water suppliers, and inefficient resource allocation for problem-solving. Furthermore,

consumers have no direct visibility into the quality of their water, leading to reliance on hearsay or expensive personal water testing kits

The proposed Aqua-sense Water Quality Monitoring System directly addresses these critical gaps by providing a real-time, multi-parameter, and transparent solution. By leveraging a network of IoT sensors, the system continuously monitors key water quality parameters such as pH, turbidity, temperature, dissolved oxygen, and conductivity. This data is transmitted wirelessly via LoRa communication to a cloud platform, where it is visualized and analyzed.

The integration of an AI-based prediction model enables the early classification of water quality and detection of potential impurities, moving from reactive to proactive management. The mobile application provides authorities with intelligent alerts and a dashboard for informed decision-making, while also offering a channel for public transparency and feedback. This system ensures continuous oversight, reduces the risk of waterborne diseases, and builds a reliable, data-driven framework for modern water quality management.

PROJECT PLAN

The Aqua-sense project will be developed using a structured, phase-driven approach to ensure the delivery of a robust and reliable water quality monitoring system. The development lifecycle encompasses requirement analysis, system design, hardware and software prototyping, integrated testing, and deployment. The project integrates ESP32 microcontrollers, a suite of water quality sensors, LoRa communication for long-range data transmission, a Firebase cloud backend, and a Flutter-based mobile application with an AI model. This plan outlines the key stages to seamlessly combine these IoT, cloud, and AI components into a functional prototype that provides real-time, actionable insights into water quality.

Planning and Setup

- Define system objectives: real-time multi-parameter water quality monitoring, long-range LoRa-based data transmission, cloud data storage and visualization, an SD card–based offline data storage system, AI-powered quality prediction, Explainable AI–based recommendation system for contamination diagnosis and corrective guidance, feedback system for the user and a user-friendly mobile application for authorities and consumers.

- Select and configure the technology stack: Arduino IDE for ESP32 firmware, Python for AI model development, Flutter for the cross-platform mobile app, and Firebase as the cloud backend and database.

- Finalize the list of sensors (pH, Turbidity, TDS, etc.) and procure all necessary hardware components, including ESP32 boards and LoRa transceivers.

- Establish the project timeline, defining milestones for hardware assembly, sensor calibration, software module completion, and integration phases.

Data Collection and Sensor Integration

- Assemble the sensor node by interfacing all water quality sensors (pH,

turbidity, Temperature, TDS, etc.) with the ESP32 microcontroller.

- Develop and upload firmware to read analog/digital signals from each sensor, convert them to meaningful units (e.g., pH value, NTU for turbidity), and package the data.
- Integrate the LoRa module with the sensor node ESP32 and configure it as a transmitter.
- Setup the base station with a second ESP32 and LoRa receiver to collect data from multiple sensor nodes.
- Calibrate each sensor using standard solutions to ensure accurate and reliable readings.

System Development

1. Embedded Module: Develop ESP32 firmware in Embedded C to handle IR sensor input, gate control via servo, OLED display output, and BLE communication.
2. Mobile Application: Build a Flutter-based app with features for:
    - Live slot availability and booking.
    - QR-code based demo payment.
    - Indoor navigation guidance using beacon data.
    - Admin dashboard for monitoring slots, payments, and usage.
3. Cloud Integration: Set up a Firebase project with Firestore/Realtime Database to store incoming sensor readings. Develop cloud functions if needed for data aggregation and implement secure access rules.

System Architecture and Integration

- Implement the full data pipeline: Sensor Node → LoRa Transmission → Base Station (LoRa Receiver with SD Card Backup) → WiFi/Ethernet → Firebase Cloud → Mobile Application.
- Incorporate an SD card–based offline data storage system at the base station

to ensure continuous data logging during LoRa or network failures. Once connectivity is restored, the stored data is automatically synchronized with the base station and uploaded to the Firebase cloud.

- Establish a robust communication protocol for LoRa packets to ensure data integrity, source identification, and reliable retransmission after recovery.
- Integrate the trained CNN-LSTM model into the Flutter app using TensorFlow Lite to enable real-time predictions on the sensor data fetched from Firebase.
- Develop intelligent alert mechanisms within the mobile application to notify users when water quality parameters exceed safe thresholds or when the AI model predicts “poor” or unsafe conditions.
- Integrate an Explainable AI (XAI)-based recommendation system that provides interpretability to AI predictions by identifying which parameters (such as turbidity, pH, or TDS) caused the water to be classified as unsafe. The system also generates actionable recommendations, guiding users and authorities on corrective measures—such as filtration, aeration, or chemical treatment—to restore water to a safe and consumable state.

User Interface Development

- Design an intuitive and clean UI for the Flutter mobile application.
- For the Admin Dashboard, create dedicated pages for Live Sensor Readings (gauges/values), Prediction Results (clear quality indicators), and Historical Data (interactive charts and graphs).
- For the End-User View, design a simple, glanceable screen showing the current water quality status (e.g., "Safe" or "Unsafe") and key parameter values.
- Implement a navigation structure that allows easy switching between different features like the dashboard, feedback form, and notification history.

Testing and Optimization

- Conduct functional testing of each sensor node and the LoRa communication link for range and reliability.
- Validate the accuracy of the AI model's predictions against known test

cases and calibrate the model if necessary.

- Perform end-to-end testing of the entire system, from sensor reading to mobile app display, to ensure data integrity and system stability.
- Test the mobile application on different devices and operating systems to ensure compatibility and performance.
- Optimize the power consumption of the sensor nodes for longer battery life in field deployments.

Deployment and Maintenance

- Deploy the cloud infrastructure on Firebase, ensuring secure and scalable data handling.
- Create a plan for the physical deployment of sensor nodes and base stations in a real- world pilot location (e.g., a water tank or distribution line).
- Set up monitoring and alerting for system errors, such as sensor node disconnections or Firebase downtime.
- Establish a maintenance schedule for periodic sensor calibration, hardware checks, and software updates for the mobile app and embedded firmware.

BUDGET AND RESOURCE ALLOCATION

The Aqua-sense Water Quality Monitoring System will be developed using a cost-effective approach, leveraging open-source technologies and a modular hardware design. The total implementation duration is estimated to be 120-150 days, covering planning, hardware prototyping, software development, AI model training, integrated testing, and pilot deployment.

- **Software and Tools**
  - Cost: ₹0 (all open-source tools).
- **Tools Used:** Visual Studio Code, Arduino IDE, Anaconda (for Python and AI model development), Flutter SDK, Firebase Console.
- **Languages:** Python (AI Model), Embedded C (ESP32), Dart (Flutter).
- **Hardware Components**
  - Cost: ₹7,000 – ₹10,000 (approx.).
  - Items Required:
    1. ESP32 Microcontrollers (x2 for sensor node and base station)
    2. LoRa Transceiver Modules (x2)
    3. Water Quality Sensors (pH, Turbidity, Temperature, TDS, Dissolved Oxygen, ORP, Electrical Conductivity)
    4. Breadboards, Jumper Wires, and Power Supplies (Battery packs for portability)
    5. Resistors and Voltage Regulators for sensor interfacing

- **Data**
  - Cost: ₹0
  - Source: Publicly available water quality datasets (e.g., from Kaggle, government repositories) in CSV/Excel format will be used for training the initial AI model. Post- deployment, real-time data will be collected from the sensors and stored in Firebase.
- **Human Resources**
  - Cost: Time and effort of team members.
  - Team Roles and Responsibilities:
    - Hardware Integration: Sensor interfacing with ESP32, LoRa communication setup, and circuit design.
    - AI/ML Development: Data preprocessing, CNN-LSTM model training, evaluation, and conversion to TensorFlow Lite for mobile deployment.
    - Mobile & Cloud Development: Flutter app development, Firebase backend setup, and cloud-to-app data integration.
    - Testing & Deployment: System integration testing, sensor calibration, and pilot deployment management.
- **Team Roles**
  - Hardware integration
  - Mobile app development
  - Cloud/database development
  - Testing and deployment

RISK ASSESSMENT

POTENTIAL RISK	MITIGATION STRATEGY
Sensor Malfunction or Drift leading to inaccurate water quality readings.	Implement periodic manual calibration using standard solutions. Design the software to flag sensor readings that deviate significantly from expected ranges for investigation.
LoRa Communication Failure between sensor nodes and the base station.	Use robust LoRa libraries with error-checking. Implement a "heartbeat" signal to monitor node connectivity and use mesh networking principles where possible to extend range.
Cloud (Firebase) Service Downtime or network issues at the base station.	Implement local data logging on the base station ESP32 (e.g., on an SD card) to cache readings during outages, with automatic syncing once the connection is restored.
AI Model Performance is Inaccurate or Unreliable with real-world data.	Continuously validate the model against lab results. Have a fallback rule-based system (e.g., "if pH > 8.5, flag as unsafe") and plan for model re-training with newly collected field data.
Power Failure at remote sensor nodes or the base station.	Utilize solar panels with battery backups for field-deployed nodes and ensure the base station is connected to an Uninterruptible Power Supply (UPS).
Data Security Breach or unauthorized access to the cloud database.	Implement strong Firebase Authentication and define strict database security rules to ensure users can only access data they are authorized to see.
Hardware Damage to sensors or controllers from environmental factors (water, dust).	House all electronics in waterproof and dustproof (IP67-rated) enclosures. Use protective circuits to guard against power surges.
Poor Mobile Application Performance or complex user interface leading to low adoption.	Follow Flutter best practices for performance. Conduct usability testing with potential users during development and iterate based on feedback to ensure an intuitive interface.
SD Card Failure or Data Corruption during Offline Storage.	Use high-endurance SD cards with periodic verification and checksum validation to prevent data loss, and enable automatic retries during synchronization.